

Sachin MB Gautham

Postdoctoral Researcher — Computational Soft Matter Physics & Machine Learning

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SUMMARY

Interdisciplinary researcher with a Ph.D. in Chemical Engineering from IIT Madras and current postdoctoral experience at IISc Bengaluru. Expert in statistical thermodynamics and soft matter physics, with a specialized focus on multi-scale molecular modeling and machine learning. Proven expertise in utilizing exascale computing to solve complex challenges in polymer science and nanocomposite systems. I seek to establish an independent, interdisciplinary research laboratory that integrates molecular theory, high-fidelity simulations, and AI-driven optimization to accelerate the design of functional nanomaterials. Committed to academic excellence through high-impact research, securing external funding, and fostering a rigorous pedagogical environment.

EDUCATION

- **Ph.D., Chemical Engineering** — Indian Institute of Technology (IIT) Madras Feb 2026
- **M.E., Chemical Engineering** — Birla Institute of Technology and Science (BITS) Pilani May 2019
- **B.Tech., Chemical Engineering** — National Institute of Technology (NIT) Karnataka, Surathkal Jul 2004

AWARDS & FELLOWSHIPS

- **ANRF National Post-Doctoral Fellowship (N-PDF)**, Anusandhan National Research Foundation (ANRF), Govt. of India — a competitive 2-year postdoctoral fellowship hosted by Prof. P. K. Maiti, Department of Physics, IISc Bengaluru (2026–2028).
- Won **First Prize for Best Poster Presentation** at PCCP-SPHERE 2025 (IISc) for the work titled “*Thickening of Liquids Using Copolymer Grafted Nanoparticles.*”
- Awarded the **International Travel Scheme (ITS) Grant** by the DST, Govt. of India, to present doctoral research at the 12th International Liquid Matter Conference (2024) at the Max Planck Institute for Polymer Research, Mainz, Germany.
- Secured a high-performance computing allocation of **1,000,000 core-hours** (CNM 83298) at Argonne National Laboratory (Carbon Cluster), USA, for large-scale simulations of nanomaterial assemblies (2023).
- Recipient of a competitive **Institute Travel Grant** from IIT Madras to present Ph.D. research at a specialized materials symposium in the USA (2023).
- Awarded **First Prize for Best Paper Presentation** at ChemPlus 2023 for developing a deep learning framework titled “*Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning.*”
- Completed Professional Industrial Training at JSW Steel (Hospet), gaining hands-on exposure to syn-gas production technologies and smelter furnace operations (2014).

RESEARCH EXPERIENCE

Research Associate–I, Indian Institute of Science, Bengaluru, India

Jan 2026 – Present

Advanced Atmospheric Water Harvesting (AWH) using Metal–Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs)

- Research conducted under the **ANRF National Post-Doctoral Fellowship (N-PDF)**, hosted by Prof. P. K. Maiti, Department of Physics, IISc Bengaluru (2026–2028).
- Served as a lead computational researcher on a high-impact, industrial-scale water harvesting project centered on CALF-20 and specialized COF architectures.
- Executed extensive Grand Canonical Monte Carlo (GCMC) simulations to rigorously evaluate multi-cycle water adsorption isotherms and hydrolytic stability of candidate MOF/COF structures.

- Developing Machine Learning Potentials (MLPs) to overcome the computational bottleneck of traditional force-field methods, accelerating water-uptake prediction speeds by orders of magnitude without sacrificing atomistic accuracy.
- Analyzed complex simulation datasets to derive structure–property relationships, directly informing the optimization of water capture efficiency and regeneration energy for industrial-scale deployment.
- Bridged the gap between fundamental molecular theory and engineering application through high-fidelity performance modeling and active participation in cross-functional technical discussions with experimental partners.

Postdoctoral Fellow, Accenture–IISc Bengaluru, India

Jul 2025 – Dec 2025

IISc–Accenture Collaborative Initiative on Predictive Analytics for Polymeric Coatings

- Spearheaded a collaborative research effort between IISc and Accenture to develop a predictive machine learning framework for the next generation of industrial paint formulations.
- Developed high-fidelity molecular models for complex ternary systems involving poly(methyl methacrylate) (PMMA), polystyrene (PS), and titanium dioxide (TiO₂) nanoparticles to investigate interfacial physics and dispersion stability.
- Architected a data-driven model to map molecular-level descriptors (polymer architecture, grafting density, and nanoparticle loading) to macroscopic rheological and mechanical properties critical to the paint industry.
- Utilized the framework to predict key performance metrics such as viscosity profiles, pigment dispersion quality, and film-forming characteristics, significantly reducing the experimental trial-and-error cycle.
- Led the translation of fundamental soft matter physics into actionable insights for industrial scale-up; ongoing optimization of nanoparticle formulations yielded promising results in enhancing coating durability and opacity.
- Oversaw technical deliverables and regular progress briefings for industrial partners; the resulting computational frameworks received positive stakeholder validation for their alignment with commercial formulation benchmarks and real-world performance requirements.

Institute Research Fellow, Indian Institute of Technology Madras, Chennai, India

Aug 2008 – Jul 2012

Deep Learning Frameworks for Potential of Mean Force (PMF) in Polymer-Grafted Nanoparticles (PGNs)

- Developed a novel deep learning architecture to map the PMF between PGNs, enabling rapid exploration of high-dimensional interaction landscapes that are traditionally computationally exhaustive.
- Engineered a robust data-generation pipeline that integrates advanced MD sampling techniques with neural networks to produce high-fidelity thermodynamic training datasets.
- Optimized deep learning models to predict interparticle interactions across a wide design space, including variations in grafting density, chain length, and solvent quality, ensuring the framework’s versatility for different systems.
- Developed a phase diagram of a polymer nanocomposite based on molecular dynamics simulations.
- Disseminated findings through high-impact peer-reviewed publications and presentations at international forums, contributing a scalable computational paradigm to the field of soft matter physics.

Nanotetrapods Promote Polymer Flow Through Confinement-Induced Packing Frustration

- Investigated anisotropic self-assembly of nanoparticles in a polymer matrix using molecular dynamics simulations.
- Developed a computational framework to model various nanoparticle geometries in nanocomposites.
- Identified the correlation between nanoparticle shapes and the rheology of nanocomposites and elucidated the molecular mechanism behind shape-dependent viscosity.
- Created a coarse-grained model for non-spherical nanoparticles and predicted their self-assembly behavior using molecular dynamics simulations.
- Collaborated with Prof. Mithun Chowdhury’s lab (IIT Bombay) to validate the coarse-grained model with experimental data.

Sequence-Defined Pareto Frontier of a Copolymer Structure

- Developed a computational framework for designing copolymer sequences using genetic algorithms and molecular dynamics simulations.
- Analyzed structure–property correlations, such as radius of gyration, from a large collection of copolymer sequences.

Nano-Enhanced Sulfonated Poly(Arylene Ether Sulfone) Composite Membranes and Their Characterization

- Designed and synthesized sPBI-SPAES with TiO₂ nanocomposite membranes for enhanced desalination properties.
- Cast and characterized reverse osmosis membranes incorporating TiO₂ nanocomposites.
- Focused on improving desalination efficiency through the integration of nanocomposites in the membrane structure.
- Conducted in-house synthesis of reverse osmosis membranes to optimize material properties.
- Enhanced mechanical properties of nanocomposite membranes compared to pristine membranes.
- Investigated the performance of TiO₂ nanocomposite membranes in reverse osmosis processes.
- Contributed significantly to publishing peer-reviewed journal articles based on the research.

INDEPENDENT RESEARCH PROJECTS

1. **Understanding Switching Dynamics of Tantalum Oxide and Titanium Oxide based Memristive Devices**
Grant No. CNM55595, Agency: Argonne National Laboratory (ANL) — Allocation: 500,000 CPU hours on the *Carbon* supercomputer (2022).
2. **Polymer Directed Self-Assembly of Polymer Grafted Nanoparticles**
Grant No. CNM 83298, Agency: Argonne National Laboratory (ANL) — Allocation: 1,000,000 CPU hours on the *Carbon* supercomputer, Center for Nanoscale Materials (2024).

MENTORSHIP ACTIVITIES

1. Mentored Mr. Abhishek for M.S. thesis on “Glass Transition in Ion-Containing Polymers” (2021–2022).
2. Mentored Ms. Sayani Karmakar for M.S. thesis on “Molecular Dynamics Simulation of Ion Transport in Polymers” (2022–2023).

TEACHING EXPERIENCE

Teaching Assistant, Indian Institute of Technology Madras

Aug 2021 – Jul 2024

- Topics included thermodynamics, unit operations laboratory, introduction to molecular simulations, statistical thermodynamics, and thermodynamics of fluid mixtures.
- Prepared course material including computer codes, lectures, homework, and practice problems.
- Led weekly laboratory and/or problem-solving and discussion sessions.
- Supervised undergraduate and graduate students in final projects and weekly homework.

MANAGEMENT & SERVICES

- Member, American Institute of Chemical Engineers (AIChE), American Physical Society (APS), and Society of Polymer Science, India (SPSI).
- Member, Organizing Committee, ChemPlus school-cum-symposium, IIT Madras, Chennai, India (2022).

JOURNAL PUBLICATIONS

1. Sarkar J, **Gautham S M B**, Ali F, Madhusudanan M, Yadav H, Datta A, Patra T K, Chandran S, and Chowdhury M. Nanotetrapods Promote Polymer Flow Through Confinement-Induced Packing Frustration. *Nature Communications* **16**, 8550 (2025).
2. Adhya P, **Gautham S M B**, Patra T K, Kaushal M, and Mondal T. Thickening of Liquids Using Copolymer Grafted Nanoparticles. *Soft Matter* **21**, 4156–4164 (2025).
3. **Gautham S M B**, and Patra T K. Deep Learning Potential of Mean Force Between Polymer Grafted Nanoparticles. *Soft Matter* **18**, 7909–7916 (2022).
4. Bale A, **Gautham S M B**, and Patra T K. Sequence-Defined Pareto Frontier of a Copolymer Structure. *Journal of Polymer Science* **60**, 2100 (2022).

5. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Nano-Enhanced Sulfonated Poly(Arylene Ether Sulfone) Composite Membranes and Their Characterization. *High Performance Polymers* **34**, 1193 (2022).
6. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Blend Membranes of Sulphonated Poly(arylene ether sulphone) and Sulphonated Polybenzimidazole and Their Characterization for Desalination Applications. *Bulletin of Materials Science* **47**, 1 (2024).
7. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Desalination Characteristics of New Blend Membranes Based on Sulfonated Polybenzimidazole and Sulfonated Poly(Arylene Ether Sulfone). *Polymer Bulletin* **80**, 7805–7824 (2023).

MANUSCRIPTS UNDER PREPARATION AND REVIEW

1. **Gautham S M B**, Banerjee R, and Maiti P K. Optimizing Water Harvesting in TTBA COF: An Advanced NVT+W Approach (2026).
2. **Gautham S M B**, and Maiti P K. Tuning the Viscoelasticity of Paint Formulations using Copolymer Grafted Nanocomposites (2026).
3. **Gautham S M B**, Banerjee R, and Maiti P K. Molecular Mechanisms of Water Adsorption in Functionalized Covalent Organic Frameworks (2027).
4. **Gautham S M B**, and Patra T K. Machine Learning Guided Design of Multi-Component Copolymer Nanocomposites for Industrial Coatings (2027).

THESIS

1. **Ph.D. Thesis**: Microstructure–Rheology Correlations in Polymer Nanocomposites, Indian Institute of Technology Madras (2026).
2. **M.Tech. Thesis**: The Synthesis of Polymer Nanocomposite Membranes for Desalination, Birla Institute of Technology and Science, Pilani (2019).
3. **B.Tech. Thesis**: Preparation and Characterization of Immobilized Bacteria in Sodium Alginate, National Institute of Technology Karnataka, Surathkal (2017).

TALKS AND POSTER PRESENTATIONS

1. **Gautham S M B**, and Patra T K. Thickening of Liquids Using Copolymer Grafted Nanoparticles. RSC Physical Chemistry Chemical Physics (PCCP) SPHERE, IISc Bengaluru, India (October 2025).
2. **Gautham S M B**, and Patra T K. Effects of Nanoparticle Shape on the Rheology of Polymer Nanocomposites. Liquid Matter Conference (LMC), Mainz, Germany (September 22–27, 2024).
3. **Gautham S M B**, and Patra T K. Non-linear Rheology of Polymer Grafted Nanoparticle Solutions. Complex Fluids (CompFlu), Chennai, India (December 2023).
4. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. Sustainable Polymer Society of India (SPSI) MACRO, Guwahati, India (December 2023).
5. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. CECAM MD60, Bengaluru, India (September 2023).
6. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. ChemPlus, IIT Madras, Chennai, India (March 2023).
7. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. American Physical Society (APS) March Meeting, Las Vegas, Nevada, USA (March 5–10, 2023).
8. **Gautham S M B**, and Patra T K. Predicting GNPs Assembly using Deep Learning. Complex Fluids (CompFlu), Kolkata, India (December 19–21, 2022).